Gyuseong Lee

PH.D STUDENT, KAIST

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Summary

Computer vision and generative AI researcher with industry experience at LG Electronics. Co-authored papers published at CVPR, ICCV (100+ citations), ICLR (oral), ECCV, and WACV, with over+400 total citations. Co-authored papers published at CVPR, ICCV, ICLR (Oral), ECCV, and WACV, with over 400 total citations, including an ICCV paper with 200+ citations. Recognized as Outstanding Reviewer at ECCV 2024.

Educations

Korea Advanced Institute of Science and Technology (KAIST) *Seoul, South Korea*

PH.D. STUDENT, KIM JAECHUL GRADUATE SCHOOL OF AI *Sep. 2026 – Present*

- Advisor: Prof. Seungryong Kim

Korea University *Seoul, South Korea*

M.S. IN COMPUTER SCIENCE AND ENGINEERING *Mar. 2022 – Feb. 2024*

- Advisor: Prof. Seungryong Kim

Korea University *Seoul, South Korea*

B.S. IN ELECTRICAL ENGINEERING *Mar. 2016 – Feb. 2022*

Experience

LG Electronics *Seoul, S.Korea*

COMPUTER VISION RESEARCH SCIENTIST *Feb. 2024 - Aug. 2026*

- Built visual synthetic data generation pipelines.
- Worked on large multimodal models (LMMs) for visual perception tasks using synthetic data.
- Developed internal open-vocabulary detection models that matched the performance of SOTA models.

Research Interests

- Diffusion-based Generative Models: Guidance Methods, Controllability, and Data Generation
- Robust Fine-tuning of Foundation Models
- Semi- & Self-Supervised Learning
- Visual Correspondence

Publications

Domain Generalization Using Large Pretrained Models With Mixture-of-Adapters *WACV 2025*

GYUSEONG LEE*, WOOSEOK JANG*, JIN HYEON KIM, JAEWOO JUNG, SEUNGRYONG KIM

- **Topics**: Domain Generalization, Robust Fine-tuning

Diffusion Model for Dense Matching *ICLR 2024 Oral Presentation*

JISU NAM, **GYUSEONG LEE**, SUNWOO KIM, HYEONSU KIM, HYOUNGWON CHO, SEYEON KIM, SEUNGRYONG KIM

- **Topics**: Diffusion Generative Models, Image Matching

Improving Sample Quality of Diffusion Models Using Self-Attention Guidance *ICCV 2023*

SUSUNG HONG, **GYUSEONG LEE**, WOOSEOK JANG, SEUNGRYONG KIM

- **Topics**: Diffusion Generative Models, Classifier-Free Guidance

* Equal contribution.

Depth-Aware Guidance with Self-Estimated Depth Representations of Diffusion Models *Pattern Recognition*
Gyeongnyeon Kim*, Wooseok Jang*, **Gyuseong Lee***, Susung Hong, Junyoung Seo, Seungryong Kim
- **Topics**: Diffusion Generative Models, Sampling Guidance, Diffusion Features

MIDMs: Matching Interleaved Diffusion Models for Exemplar-based Image Translation *AAAI 2023*
Junyoung Seo*, **Gyuseong Lee***, Seokju Cho, Jiyoung Lee, Seungryong Kim (Co-first author)
- **Topics**: Diffusion Generative Models, Image-to-Image Translation, Image Matching

Towards Flexible Inductive Bias via Progressive Reparameterization Scheduling *ECCV 2022 VIPriors Workshop*
Yunsung Lee*, **Gyuseong Lee***, Kwangrok Ryoo*, Hyojun Go*, Jihye Park*, Seungryong Kim
- **Topics**: Model Architecture, Vision Transformer

ConMatch: Semi-supervised Learning with Confidence-Guided Consistency Regularization *ECCV 2022*
Jiwon Kim, Youngjo Min, Daehwan Kim, **Gyuseong Lee**, Junyoung Seo, Kwangrok Ryoo, Seungryong Kim
- **Topics**: Image Classification, Semi-Supervised Learning

Semi-Supervised Learning of Semantic Correspondence with Pseudo-Labels *CVPR 2022*
Jiwon Kim*, Kwangrok Ryoo*, Junyoung Seo*, **Gyuseong Lee***, Daehwan Kim, Hansang Cho, Seungryong Kim (Co-first author)
- **Topics**: Image Matching, Semi-Supervised Learning

AggMatch: Aggregating Pseudo Labels for Semi-Supervised Learning *arXiv preprint*
Jiwon Kim, Kwangrok Ryoo, **Gyuseong Lee**, Seokju Cho, Junyoung Seo, Daehwan Kim, Hansang Cho, Seungryong Kim *Jan. 2022*
- **Topics**: Image Classification, Semi-Supervised Learning

Awards & Scholarships

2024	**Outstanding Reviewer**, ECCV 2024	*Milano, Italy*
2023	**Best Paper Award**, 33th Artificial Intelligence and Signal Processing, IEIE	*Seoul, S.Korea*
2022	**Best Paper Award**, 32nd Joint Conference of Signal Processing, IEIE	
2020	**Dean's List**, Korea University	
2021	**Semester High Honors**, Korea University	
	National Science & Technology Scholarship, Korea Student Aid Foundation	
2017	**Semester High Honors**, Korea University	
2016	**Semester High Honors**, Korea University	

Skills

Core Python PyTorch Docker Git LaTeX Gerrit
Previously used C MATLAB Verilog

Projects

On-device Scene Understanding & Gesture Interaction for Pro-active Home Agents *LG Electronics (PoC)*
Feb. 2025 – Mar. 2025
- Prototyped a Raspberry Pi 5 home agent featuring motion detection, image captioning, and LLM-based scene-state recognition; defined three agent states (lying, sitting, fallen) and implemented real-time hand-gesture control for state transitions.

Multi-domain Generalizable Anomaly Classification (Industry Project) *Samsung Electro-Mechanics*
Sep. 2022 - Apr. 2023
- Investigated OOD robustness and model bias; developed domain generalization and label-efficient adaptation leveraging large pretrained models for anomaly classification.

* Equal contribution.

Research Infrastructure & GPU Monitoring System *Korea University*

Sep. 2022 - Dec. 2023

- Developed a centralized dashboard to monitor health and resource utilization across the lab's multi-GPU servers using DCGM-Exporter, Prometheus/Grafana, and Uptime-Kuma to address hardware issues.

Semi-supervised & continual learning for anomaly classification (Industry Project) *Samsung Electro-Mechanics*

July. 2021 - May. 2022

- Utilized a contrastive-based loss function to improve the classification accuracy of fine-grained, long-tailed defective samples in a semi-supervised setting.
- Applied knowledge distillation to preserve model performance over time in continual learning scenarios.

Peer Review

- Selected as an **outstanding reviewer at ECCV 2024**
- Served as a reviewer for top-tier conferences including NeurIPS, CVPR, ICCV/ECCV, ICLR, AAAI, and WACV.